

INDEPENDENT SQL PORTFOLIO PROJECT

Brazilian E-Commerce Analytics

Order Trends, Payments, Freight & Delivery Performance using SQL

Problem statement

Using the Brazilian E-Commerce Public Dataset by Olist, analyze order and transaction data to uncover useful insights about purchase trends, customer geography, payment behavior, item prices, freight costs, and delivery performance. The goal is to answer defined business questions using SQL, validate the calculations, summarize measurable findings, and identify areas where further operational investigation may be useful.

Project objective

Build a repeatable SQL analysis across related CSV tables. Profile the data first, aggregate at the correct grain, compare trends over time and across Brazilian states, evaluate payment and installment patterns, and examine actual delivery outcomes against estimates. Present the results with clear tables, charts, business interpretations, and documented limitations.

Expected deliverables & scope

The project is organised into six original analysis areas and six supplemental portfolio questions.

Expected deliverables

Deliverable	What it contains
SQL analysis	Queries for 17 original requests plus six supplemental portfolio questions.
Business findings	Numeric insights and caveats for trends, geography, payments, prices, freight, delivery and reviews.
Logical data model	Table grain, key relationships and join cautions.
GitHub README	Project overview, SQL methods, numeric results, charts and limitations.

Data context

The supplied CSV files contain 99,441 orders, 112,650 order items, 103,886 payment rows, and customer records across 27 state codes and 4,119 cities. Purchase timestamps run from 2016-09-04 21:15:19 through 2018-10-17 17:30:18.

Exploratory data analysis - Inspect source data types, purchase-date coverage, and city/state counts.

Order trends - Compare annual growth, monthly patterns and time-of-day order activity.

Regional evolution - Analyze state-by-month order counts and customer concentration.

Money movement - Compare Jan-Aug payment value across 2017/2018 and state-level item price / freight.

Delivery operations - Calculate delivery duration, estimate variance and state rankings.

Payment behavior - Compare monthly order counts by payment type and installment count.

Portfolio extensions - Add status, payment-value mix, delivery timeliness, reviews, product category value and weekday patterns.

Method controls

Count distinct order_id when counting orders. Keep payment rows separate from order-item rows when summing money. Exclude undelivered orders from delivery-duration averages. For delivery variance, actual minus estimated means negative = early and positive = late. Treat incomplete year coverage cautiously.

Project positioning: independent portfolio analysis using public Olist data; not affiliated with Target.